22-2 VLAN and Inter-VLAN Routing Troubleshooting - Answer Key

In this lab you will troubleshoot VLANs.

VLAN Troubleshooting

1) The campus LAN has been configured but users are complaining of connectivity issues.

   Your task is to troubleshoot the configuration and ensure that all PCs have connectivity to each other. The exercise is complete when you can ping from the Eng1 PC to Sales1.

   There is a configuration error on each switch. You will need to find and fix all errors before the network will work properly.

   R1 will be used as the default gateway. The PCs and router are configured correctly. Do not edit the configuration on the PCs or router.

   Note that some commands may be disabled during the CCNA exam so you should know the different troubleshooting commands.

   You must not use the 'show running-configuration' command (or any derivative of it such as 'show run | begin vlan) in this exercise.

   You must not change the VTP configuration on any switch.

   Packet Tracer can be a little slow to update after fixing problems. Try pings twice when verifying connectivity.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

First check connectivity between PCs in the same VLAN. Ping from the Eng1 PC to Eng3.

```
C:\>ping 10.10.10.12

Pinging 10.10.10.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

The ping fails. Let's check the VLANs are present on all the switches.

```
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   Eng                              active    Fa0/2
20   Sales                            active    Fa0/1, Fa0/3
199  Native                           active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
SW2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   Eng                              active    Fa0/1
20   Sales                            active    Fa0/2
199  Native                           active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
SW3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
199  Native                           active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

The Eng and Sales VLANs are missing on SW3. Attempt to add the VLANs.

```
SW3(config)#vlan 10
VTP VLAN configuration not allowed when device is in CLIENT mode.
```

This fails because the switch is a VTP client. Check the VTP configuration on the other switches.

```
SW1#show vtp status
VTP Version : 2
Configuration Revision : 7
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
VTP Operating Mode : Server
VTP Domain Name : Flackbox
VTP Pruning Mode : Disabled
VTP V2 Mode : Disabled
VTP Traps Generation : Disabled
MD5 digest : 0xEE 0x08 0x89 0x96 0xBC 0xE5 0x0F 0xA9
Configuration last modified by 0.0.0.0 at 3-2-93 00:27:15
Local updater ID is 0.0.0.0 (no valid interface found)
```

```
SW2#show vtp status
VTP Version : 2
Configuration Revision : 0
Maximum VLANs supported locally : 1005
Number of existing VLANs : 8
VTP Operating Mode : Transparent
VTP Domain Name :
VTP Pruning Mode : Disabled
VTP V2 Mode : Disabled
VTP Traps Generation : Disabled
MD5 digest : 0x7D 0x5A 0xA6 0x0E 0x9A 0x72 0xA0 0x3A
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
```

SW3 is failing to synchronise its VLAN database with the VTP Server SW1. We need to check the trunks from SW1 to SW3.

Check the configuration on the trunk from SW1 to SW2.

```
SW1#show interface g0/1 switchport
Name: Gig0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 199 (Native)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More-- |
```

It is configured as a trunk with native VLAN 199. Check the configuration matches on SW2.

```
SW2#show interface g0/1 switchport
Name: Gig0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 199
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More-- |
```

That looks okay. Check the configuration on the trunk from SW2 to SW3.

```
SW2>en
SW2#show interface g0/2 switchport
Name: Gig0/2
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 199
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More--
```

It is also configured as a trunk with native VLAN 199. Check the configuration matches on SW3.

```
SW3#show int g0/2 switchport
Name: Gig0/2
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More--
```

There's the problem, it's configured as an access port in VLAN 1. Configure it as a trunk port with native VLAN 199.

```
SW3(config)#int g0/2
SW3(config-if)#switchport mode trunk
SW3(config-if)#switchport trunk native vlan 199
```

Check to see if SW3 learns the VLANs from the VTP Server SW1 now.

```
SW3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1
10   Eng                              active    Fa0/3
20   Sales                            active    Fa0/1, Fa0/2
199  Native                           active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

That's better. Check to see if Eng1 can ping Eng3 now.

```
C:\>ping 10.10.10.12

Pinging 10.10.10.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

The ping still fails. Check the PCs are in the correct VLANs.

```
SW3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1
10   Eng                              active    Fa0/3
20   Sales                            active    Fa0/1, Fa0/2
199  Native                           active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

On SW3, FastEthernet 0/1 (Sales1) and 0/2 (Sales2) are in the Sales VLAN, FastEthernet 0/3 (Eng3) is in the Eng VLAN. That looks okay.

```
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   Eng                              active    Fa0/2
20   Sales                            active    Fa0/1, Fa0/3
199  Native                           active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

On SW1, FastEthernet 0/1 (Eng1) and 0/3 (Sales2) are in the Sales VLAN, FastEthernet 0/2 (Eng2) is in the Eng VLAN. FastEthernet 0/1 has been configured with the wrong VLAN. We can also verify this at the interface level.

```
SW1#show interface f0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 20 (Sales)
Trunking Native Mode VLAN: 1 (default)
```

It has been configured as an access port but in the Sales VLAN 20. Change this to the Eng VLAN 10.

```
SW1(config)#int f0/1
SW1(config-if)#switchport access vlan 10
```

Try the ping from Eng1 to Eng3 again.

```
C:\>ping 10.10.10.12

Pinging 10.10.10.12 with 32 bytes of data:

Reply from 10.10.10.12: bytes=32 time=14ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time=13ms TTL=128
Reply from 10.10.10.12: bytes=32 time=22ms TTL=128

Ping statistics for 10.10.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 22ms, Average = 12ms
```

We now have layer 2 connectivity across the LAN.

Check connectivity from the Eng1 PC to Sales1.

```
C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.10: bytes=32 time=15ms TTL=127
Reply from 10.10.20.10: bytes=32 time=14ms TTL=127
Reply from 10.10.20.10: bytes=32 time=13ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 15ms, Average = 14ms
```

The ping fails. We know layer 2 connectivity is good so this must be an inter-VLAN routing problem. Check the router on a stick configuration on R1.

```
R1#show ip int brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 unassigned YES unset up up
FastEthernet0/0.10 10.10.10.1 YES manual up up
FastEthernet0/0.20 10.10.20.1 YES manual up up
FastEthernet0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down

R1#show int f0/0.10
FastEthernet0/0.10 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4a79.3401 (bia
0002.4a79.3401)
Internet address is 10.10.10.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 10
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never

R1#show int f0/0.20
FastEthernet0/0.20 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4a79.3401 (bia
0002.4a79.3401)
Internet address is 10.10.20.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 20
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
```

That all looks okay. From the topology diagram, interface FastEthernet 0/1 on SW2 is connected to FastEthernet 0/0 on R1. We can verify this with CDP.

```
SW2#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce    Holdtme    Capability    Platform    Port ID
SW1          Gig 0/1           143           S         2960        Gig 0/1
R1           Fas 0/1           143           R         C2800       Fas 0/0
SW3          Gig 0/2           165           S         2960        Gig 0/2
```

Check the configuration on interface FastEthernet 0/1 on SW2.

```
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (Eng)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
--More—
```

It's been configured as an access port in Eng VLAN 10. For router on a stick it should be configured as a trunk port. Correct this.

```
SW2(config)#int f0/1
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport trunk encapsulation dot1q
```

Try to ping from the Eng1 PC to Sales1 again.

```
C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time=13ms TTL=127
Reply from 10.10.20.10: bytes=32 time=36ms TTL=127
Reply from 10.10.20.10: bytes=32 time=10ms TTL=127
Reply from 10.10.20.10: bytes=32 time=11ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 36ms, Average = 17ms
```

We now have full layer 2 and layer 3 connectivity across the LAN.

To summarise the issues, Eng1 was in the Sales VLAN on SW1, the link to R1 on SW2 was configured as an access port, as was the link from SW3 to SW2.